

Professional Summary

A motivated fourth year in MEng Mechanical Engineering student on track for a First Class Honours with strong foundation in CAD, additive manufacturing, robotics and multidisciplinary design projects. A founding member of the ORION Olympus Rover team, applying SolidWorks, FEA, and vibrations to develop competition-ready systems. Seeking a graduate role to apply analytical and problem-solving skills to real-world

Education

University of Leeds

Meng Mechanical Engineering (MEng) – Year 3

Classification: On track for a **First Class Honours**

Relevant modules: Thermodynamics & Fluid Mechanics; Engineering Design & CAD (SolidWorks); Electrical Circuits and Systems; Programming for Engineers (MATLAB, C++); Materials & Manufacturing Processes; Control Systems & Instrumentation; Machine intelligence and Robotics (Python)

Selected Project:

- **Individual Research Project (Year 3)** – “Novel Energy Harvesting Device: Design and Manufacturing via additive manufacturing with the use of triboelectric generators”
- **Mars Rover Design project** – Designed and manufactured a Mars rover under competition specifications in SolidWorks and C++ optimising for vibrations and speed through an obstacle course, including a mechanism to drop a payload in a designated area. Finishing in the top 8 of 35 and winning Most Innovative Design.

Alleyn’s School

Dulwich, England, June 2023

A levels: Mathematics (A*), Design and Technology (A*), Physics (A), Further Mathematics (A)

Work experience

Orion “Olympus” Rover Project

Leeds, October 2025-

Present

Mechanical Engineering Specialist & Founding Member

- Founding member of Olympus, a new division within the ORION student project, established to design and build a rover for the UKSEDS Rover Challenge 2026.
- Currently designing a rover within a multidisciplinary team of 10 students following competition rules and innovation within a sub-team of 5
- Applying CAD and simulation tools (SolidWorks, FEA, Bambu Labs) to prototype and optimise mechanical systems for performance, weight, and reliability
- Demonstrating teamwork, communication and project management skills in working with a team since its inception

Personal Projects and University Projects

A website to view my portfolio: <https://jaspertaffs.github.io/>

Hydroelectric Motor

- Designing, modelling and manufacturing a outboard motor to both harvest and produce energy whilst at sea to be providing in sailing and other marine
- Improving design, prototyping and manufacturing skills alongside theoretical knowledge to produce a working prototype

V6 SolidWorks Project

- Following 2D drawings a V6 engine parts were all made and then modelled together to provide skills within SolidWorks from being able to read 2D drawings and modelling of assemblies and motion

Cochlear implant toy

- Designed and manufactured an educational toy for hearing impaired children between the ages of 18 months and 3 years old primarily with cochlear implants, gathering primary data from a support group of 20 parents
- Conducted research within paediatric audiology focusing on early children development and the speech banana for speech within children
- Improving skills of product design, research and manufacturing whilst ensuring tenacity and creative critical thinking

SKILLS AND INTERESTS

- **Technical Skills:** MATLAB; C++ (micro-processing); Microsoft Excel; SolidWorks; CAD; Learning Python for machine learning and Robotics, Wood working, Additive manufacturing with Bambu Lab printers
Awards & Achievements: Gold Duke of Edinburgh, Julian Wolchover Design and Technology award at Alleyn's

Interests: Play Field Hockey at University, Casual Tennis and Acrylic Painting